INCH-POUND
MIL-DTL-24640/21E
8 November 2011
SUPERSEDING
MIL-DTL-24640/21D
22 August 2002

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, -20 TO +105 °C, 600 VOLTS, LIGHTWEIGHT, FOUR-CONDUCTOR, LOW POWER, TYPE FXW (INCLUDING VARIATION TYPE FXOW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24640.

Construction (Watertight with Circuit Integrity)

First	-	Copper conductor, coated or uncoated in accordance with ASTM B8, 7 strands in accordance with tables I and II for the applicable type.
Second	-	Mica glass tape insulation 0.0029±0.0002 inch (50 percent overlap maximum and minimum overlap of 45 percent). Conformance testing shall be in accordance with MIL-DTL-81381 prior to cabling.
Third	-	Opaque-pigmented polyimide tape 0.001 inch (50 percent overlap minimum).
Fourth	-	Opaque-pigmented polyimide 0.001 inch with 0.0001 inch Fluorinated Ethylene Propylene (FEP) tape (50 percent overlap minimum).
Fifth	-	Modified aromatic polyimide resin coating 0.0005 inch minimum (0.001 inch nominal). Identification codes applied using Method 1 or 3.

Construction Variant for Type FXW

Sixth	-	Four component wires shall be cabled together. Fillers may be used, as necessary, to form a round cable.
Seventh	-	Optional binder tape.
Eighth	-	Cross-linked polyolefin jacket (see table I for thickness). NOTE: Manufacturer's identification tape or thread is required (see 3.7.3).

Construction Variant for Type FXOW

Eighth	-	Braided shield, tin-coated copper strands. Braid must be constructed to meet surface transfer impedance requirements as listed herein.
Ninth	-	Optional binder tape.
Tenth	-	Cross-linked polyolefin jacket (see table II for thickness). NOTE: Manufacturer's identification tape or thread is required (see 3.7.3).

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TABLE I. Construction details.

PIN M24640/21	Type number	AWG size	Component wire diameter (inch)		Jacket thickness (inch) (min.)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max.)	Conductor resistance per 1000 feet (ohms)
			(min.)	(max.)		(min.)	(max.)		
-01UN	FXW-3	16	0.075	0.081	0.034	0.266	0.286	78.2	4.35
-02UN	FXW-4	14	0.091	0.097	0.038	0.315	0.339	115.1	2.88

TABLE II. Construction details - overall shield.

PIN M24640/21	Type number	AWG size	Component wire diameter (inch)		Jacket thickness (inch) (min.)	Cable diameter (inch)		Cable weight (pounds per 1000 feet) (max.)	Conductor resistance per 1000 feet (ohms)
			(min.)	(max.)		(min.)	(max.)		
-01UO	FXOW-3	16	0.075	0.081	0.038	0.324	0.350	131.6	4.35
-02UO	FXOW-4	14	0.091	0.097	0.041	0.366	0.394	164.9	2.88

REQUIREMENTS:

Qualification required.

Construction details: See tables I and II.

Component wire: For polyimide tape constructions, the following additional tests shall be applicable in accordance with MIL-DTL-81381.

Temperature rating	150 °C (302 °F) (max.) conductor temperature
Voltage rating	600 volts (root mean square) at sea level
Blocking	Oven temperature, 150±2 °C (302±3.6 °F) no sticking of conductors
Flammability	3 seconds (max.) after flame
	3.0 inches (max.) flame travel
	No flaming of tissue paper
Humidity resistance	5 megohms for 1000 feet (min.) insulation resistance after humidity exposure
Impulse dielectric test	8.0 kV, 100 percent impulse test
Insulation resistance	2500 megohms for 1000 feet (min.)
Polyimide cure test	No cracking
Shrinkage	0.062 inch (max.) at 180 °C
Surface resistance	5 megohms inches (min.), initial and final readings
Thermal shock	Oven temperature, 150±2 °C (302±3.6 °F)
Maximum change in measurement	0.031 inch

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Wet dielectric test	2500 volts (root mean square)

INSPECTION:

Basic Electricals:	
Conductor resistance (ohms/1000 feet at 25 °C, max.)	See tables I and II
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2000
Conductor to shield (FXOW only)	1000
Shield to shield	Not applicable
Insulation resistance (megohms/1000 feet, min.)	
Conductor to conductor	2500
Conductor to overall shield (Type FXOW only)	2500
Shield to shield (component shield only)	Not applicable
Shield to shield (component shield to overall shield)	Not applicable
Conductor and shield continuity	Not applicable
Jacket flaws (Type FXOW only)	No failure
Group A:	
Visual and dimensional	No failure
Capacitance	
Mutual capacitance (at 1 MHz, picofarads/foot (pF/ft), max.)	Not applicable
Capacitance unbalance (percent, max.)	Not applicable
Characteristic impedance (at 1 MHz, ohms)	Not applicable
Watertightness (see table IX of MIL-DTL-24640 for limits of water leakage)	No failure
Weight	No failure
Group B:	
Cold bending	No failure
Drip	No failure
Cross-link proof test (percent, max.)	
Jacket	50
Tear (lb/in thickness, min.)	35
Physicals (unaged)	
Jacket	
Tensile strength (lb/in^2, min.)	1300

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Elongation (percent, min.)	160
Attenuation (at 3 MHz, dB/100 feet, max.)	Not applicable
Gas flame (1 hour)	No failure

Group C:

Physicals (aged), air oven	
Jacket	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (conductor insulation, cycles, Method 1 only, min.)	25
Permanence of printing (jacket, cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable sealant removability	No failure
Shrinkage	No failure
Surface transfer impedance (Type FXOW only)	
Milliohms per meter (max.)	700
EMP response (dB, min.)	60

Group D:

Flame propagation	No failure

Qualification Inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable)	No failure
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50

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Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	10
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - Type A (hardness, min.)	80
Weathering (jacket)	No failure
Fungus resistance	No failure
Abrasion resistance	No failure
Gas flame (3 hour)	No failure
Component wire	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
FXW-3, FXW-4, FXW-3, FXW-4	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2009-096)

Review Activities:
Army – AR, AV, CR, CR4
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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